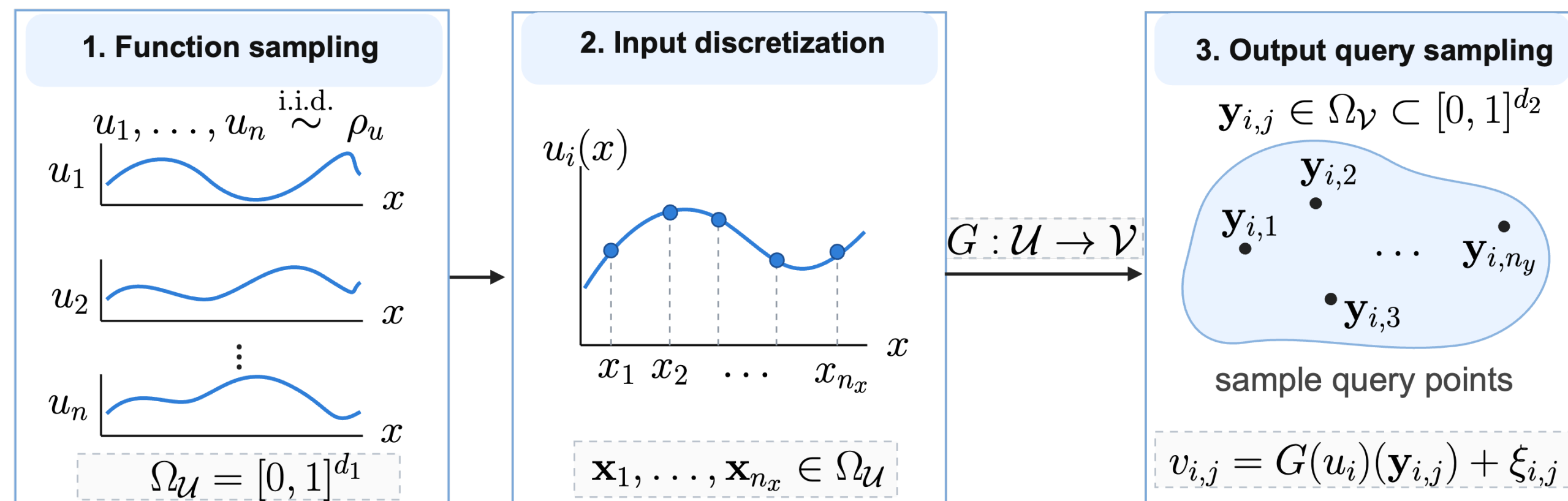




Transformer Operator Learning: Setup & Objective

We consider supervised learning of a nonlinear operator $G : \mathcal{U} \rightarrow \mathcal{V}$ from finitely sampled input functions and pointwise output queries.



The training sample can be written as $\mathcal{S} = \{(S_{n_x}(u_i), \mathbf{y}_{i,j}, v_{i,j}) : i \in [n], j \in [n_y]\}$.

For an input function u , define its discretized representation

$$S_{n_x}(u) = \{(\mathbf{x}_j, u(\mathbf{x}_j))\}_{j=1}^{n_x}.$$

The Transformer predicts

$$T_{\Theta}(S_{n_x}(u), \mathbf{y}) \approx G(u)(\mathbf{y}).$$

Statistical objective. For the empirical risk minimizer $\hat{T} \in \arg \min_{T \in \mathcal{H}} \hat{\mathcal{L}}_{\mathcal{S}}(T)$, we study

$$\mathbb{E}_{\mathcal{S}} \mathcal{L}(\hat{T}) = \mathbb{E}_{\mathcal{S}} \mathbb{E}_{u \sim \rho_u} \mathbb{E}_{\mathbf{y} \sim \rho_y} \left[\left(\hat{T}(S_{n_x}(u), \mathbf{y}) - G(u)(\mathbf{y}) \right)^2 \right]$$

Main question. How does this error scale with model size and training-data size?

Theoretical Transformer Architecture & Structural Assumptions

Given $S_{n_x}(u)$ and a query $\mathbf{y} \in \Omega_{\mathcal{V}}$, we organize the sampled input function and query into the token sequence

$$\mathbf{X}(u, \mathbf{y}) = \begin{bmatrix} \mathbf{x}_1 & \cdots & \mathbf{x}_{n_x} & \mathbf{0}_{d_1} & \cdots & \mathbf{0}_{d_1} \\ \mathbf{0}_{d_2} & \cdots & \mathbf{0}_{d_2} & \mathbf{y} & \cdots & \mathbf{0}_{d_2} \\ u(\mathbf{x}_1) & \cdots & u(\mathbf{x}_{n_x}) & 0 & \cdots & 0 \end{bmatrix}.$$

An affine embedding with structural-positional encoding gives

$$\mathbf{Z}_0 = \mathbf{W}_E \mathbf{X}(u, \mathbf{y}) + \mathbf{b}_E \mathbf{1}_P^{\top} + \mathbf{P}_{\text{pos}}.$$

The hidden sequence is then processed by two encoder blocks, followed by a linear readout:

$$\mathbf{Z}_0 \rightarrow \text{Block 1} \rightarrow \text{Block 2} \rightarrow T_{\Theta}(S_{n_x}(u), \mathbf{y}).$$

Assumption 1: Smooth and low-dimensional input class.

$$\sup_{u \in \mathcal{U}} \|u\|_{C^{\alpha}} \leq B_{\mathcal{U}}, \quad C_U \leq C_1 r_u^{-d_{\mathcal{U}}}.$$

Here C_U is the size of an r_u -cover of $\mathcal{U}$ in L^2 , that is, $\sup_{u \in \mathcal{U}} \min_{k \in [C_U]} \|u - u_k\|_{L^2(\Omega_{\mathcal{U}})} \leq r_u$.

Assumption 2: Hölder-stable target operator.

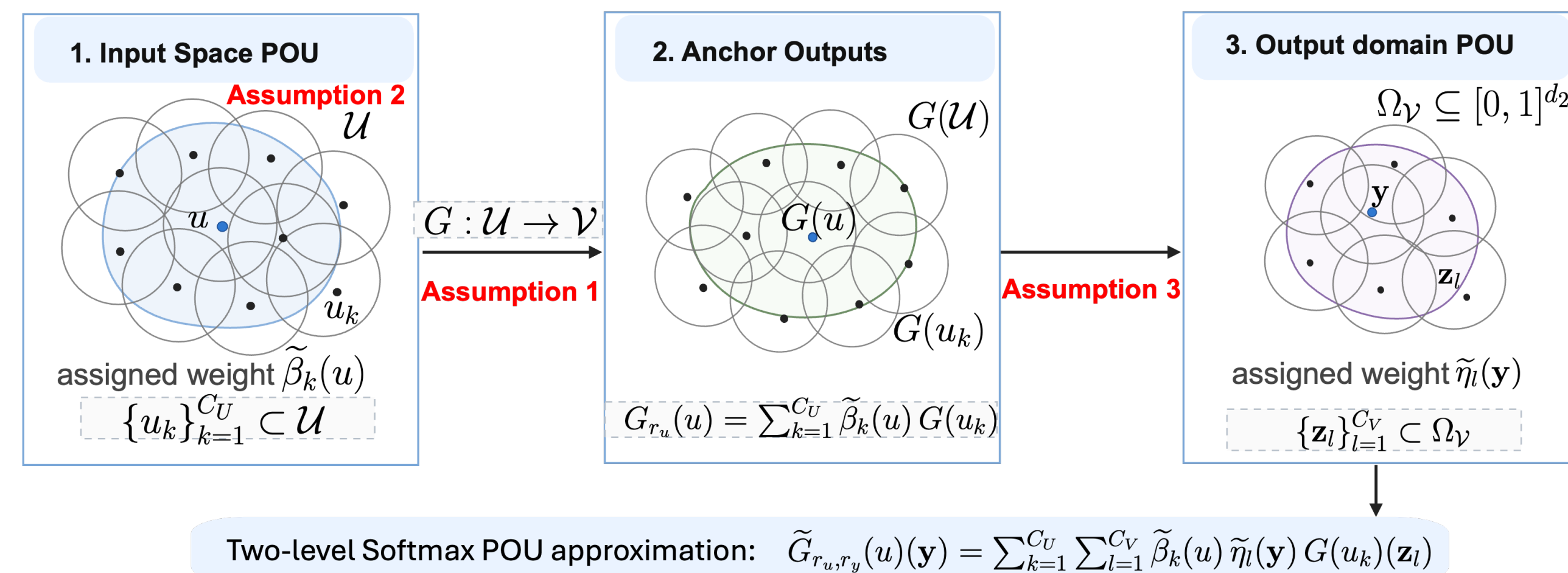
$$\|G(u_1) - G(u_2)\|_{L^2(\rho_y)} \leq L_G \|u_1 - u_2\|_{L^2(\Omega_{\mathcal{U}})}^{\gamma}.$$

Assumption 3: Smooth output class.

$$\sup_{v \in \mathcal{V}} \|v\|_{C^{\beta}} \leq B_{\mathcal{V}}.$$

Two-Level Softmax Partition of Unity

The key idea is to localize jointly in the input-function space and the output domain.



The two POU weights are

$$\tilde{\beta}_{k,n_x}(u) = \frac{\exp\left(2M_u \langle u, u_k \rangle_{n_x} - M_u \|u_k\|_{L^2}^2\right)}{\sum_{k'=1}^{C_U} \exp\left(2M_u \langle u, u_{k'} \rangle_{n_x} - M_u \|u_{k'}\|_{L^2}^2\right)}, \quad \tilde{\eta}_{\ell}(\mathbf{y}) = \frac{\exp\left(2M_y \mathbf{y}^{\top} \mathbf{z}_{\ell} - M_y \|\mathbf{z}_{\ell}\|_2^2\right)}{\sum_{\ell'=1}^{C_V} \exp\left(2M_y \mathbf{y}^{\top} \mathbf{z}_{\ell'} - M_y \|\mathbf{z}_{\ell'}\|_2^2\right)},$$

where M_u and M_y control the localization sharpness.

For each anchor pair $(u_k, \mathbf{z}_{\ell})$, Block 1 constructs the joint input-output localization logit. Block 2 applies Softmax and aggregates the corresponding anchor values $G(u_k)(\mathbf{z}_{\ell})$.

Approximation and Generalization Scaling Laws

Complexity exponent. The model and data scaling rates are governed by

$$s = \frac{d_{\mathcal{U}}}{\gamma} + \frac{d_2}{\beta} + \frac{2d_1}{\alpha} \left(1 + \frac{2}{\gamma}\right).$$

Theorem 1: Transformer approximation. For any target accuracy $\epsilon > 0$, there exists a Transformer T_{Θ} such that

$$\sup_{u \in \mathcal{U}} \|T_{\Theta}(S_{n_x}(u), \cdot) - G(u)\|_{L^2(\rho_y)} \leq \epsilon,$$

with total parameter count

$$N_{\text{total}} \lesssim \epsilon^{-s} \left(\log \epsilon^{-1}\right)^{2d_1/\alpha}.$$

Thus, up to logarithmic factors, approximation error $\sim N_{\text{total}}^{-1/s}$.

Theorem 2: Generalization of the Transformer ERM. For the empirical risk minimizer $\hat{T}_{\epsilon}$,

$$\mathbb{E}_{\mathcal{S}} \mathcal{L}(\hat{T}_{\epsilon}) \lesssim \epsilon^2 + \left(1 + \frac{\sigma^2}{n_y}\right) \frac{\epsilon^{-s} (\log \epsilon^{-1})^{2d_1/\alpha+1}}{n}.$$

Balancing approximation and estimation errors gives

$$\mathbb{E}_{\mathcal{S}} \mathcal{L}(\hat{T}_{\epsilon}) \lesssim \left[\frac{1 + \sigma^2/n_y (\log n)^{2d_1/\alpha+1}}{n} \right]^{\frac{2}{2+s}}.$$

Thus, up to logarithmic and noise-dependent factors, generalization error $\sim n^{-2/(2+s)}$.

$$d_{\mathcal{U}} \uparrow \implies s \uparrow \implies \text{slower model and data scaling.}$$

PDE Experiments: Does Intrinsic Dimension Affect Scaling?

We test the theoretical prediction $d_{\mathcal{U}} \uparrow \implies$ slower empirical data scaling.

Burgers' equation

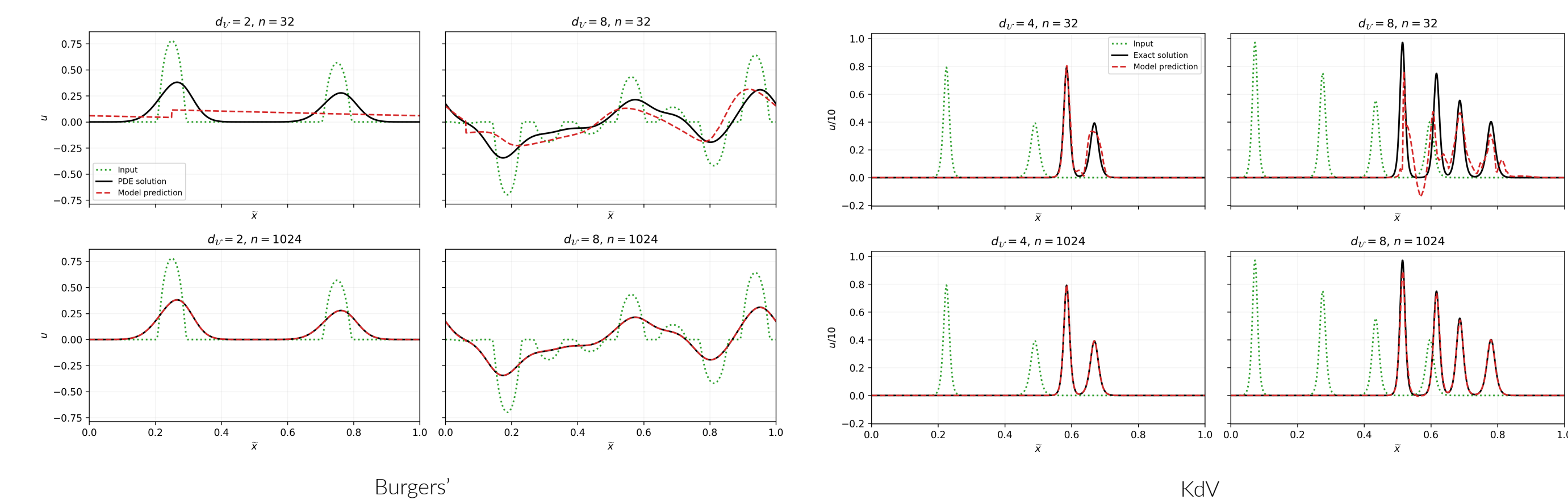
$$u_t + uu_x = \nu u_{xx}, \quad \nu = 0.02, \quad d_{\mathcal{U}} \in \{2, 8\},$$

KdV equation

$$u_t + 6uu_x + u_{xxx} = 0, \quad d_{\mathcal{U}} \in \{4, 8\}.$$

For each PDE and each $d_{\mathcal{U}}$, we train the same Transformer architecture over increasing numbers of training functions n and evaluate the test MSE on held-out functions.

Representative predictions.

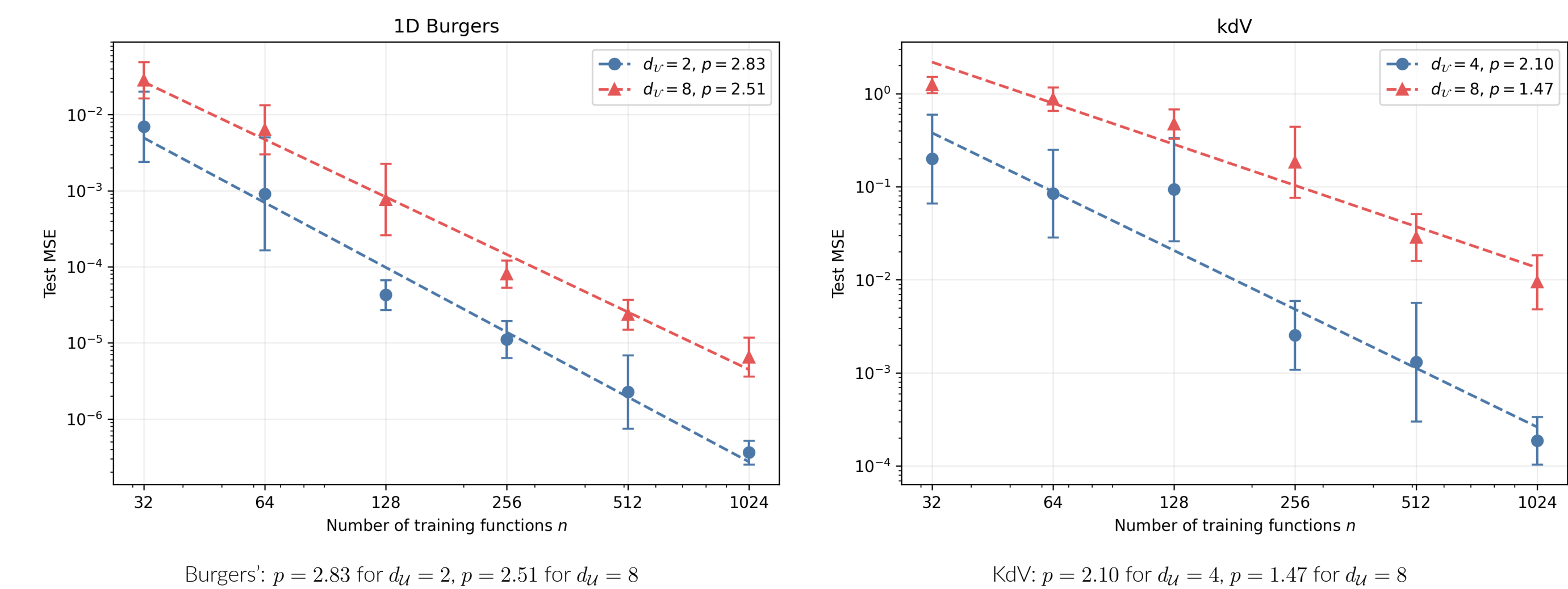


As the training-set size increases, the predicted solutions become substantially more accurate for both PDEs. This motivates quantifying the improvement through an empirical power-law fit.

Empirical data scaling. We fit the test error by

$$\text{MSE}(n) \approx C n^{-p},$$

where p is the empirical data-scaling exponent.



Key takeaway. For both PDEs, larger $d_{\mathcal{U}}$ yields a smaller empirical scaling exponent p , consistent with the predicted slower data scaling.

Selected References

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[2] H. Liu, Z. Zhang, W. Liao, and H. Schaeffer, *Neural Scaling Laws of Deep ReLU and Deep Operator Network: A Theoretical Study*, arXiv:2410.00357, 2026.

[3] H. Yan, Z. Shi, Y. Xi, P. Chen, and W. Liao, *Neural Scaling Laws of Transformer Operator Network*, arXiv:2609.33533, 2026.